

Bowen Yang

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Research Interests

Robot learning and embodied AI, with a focus on vision–language–action models, structured representations for robotic manipulation, and robust autonomous systems.

Education

Shanghai Jiao Tong University

Bachelor's studies in French; Minor in Information Engineering
SJTU Paris Elite Institute of Technology. Advisor: **Prof. Junchi Yan**.
Average score: **91.69** · Rank: **2/77**

Aug 2023 –

Jun 2027 (expected)

Publications

ELAN4D: Embodiment-Centric 4D Supervision for Vision-Language-Action Models via Plug-and-Play Adaptation 2026

Zeyuan He*, **Bowen Yang***, Zhirui Fang*, Keru Zhou*, Lei Jiang, Jingjing Qian, Fan Mo, Junchi Yan, Philip Torr, Xiu Li, Li Jiang, Jialin Yu†

Conference on Robot Learning (CoRL)

GuidedVLA: Specifying Task-Relevant Factors via Plug-and-Play Action Attention Specialization 2026

Xiaosong Jia*, **Bowen Yang***, Zuhao Ge*, Xian Nie*, Yuchen Zhou*, Cunxin Fan*, Yufeng Li, Yilin Chai, Chao Jing, Zijian Liang, Qingwen Bu, Haidong Cao, Chao Wu, Qifeng Li, Zhenjie Yang, Chenhe Zhang, Hongyang Li, Zuxuan Wu†, Junchi Yan†, Yu-Gang Jiang†

Robotics: Science and Systems (RSS)

* Equal Contribution; † Corresponding Author

Research Experience

Embodiment-Centric 4D Supervision for Robot Learning

Mar 2026 – May 2026

Co-first author · [ELAN4D](#), [CoRL 2026](#)

Research collaboration with Torr Vision Group, University of Oxford.

- Proposed using **URDF-based forward kinematics** to derive future **robot-joint and end-effector keypoint tracks** from proprioception as low-cost **4D supervision**.
- Proposed a **ControlNet-style residual branch** for 4D supervision, **isolating auxiliary gradients** from the pretrained VLM while preserving the base policy's inference interface.
- Contributed to **ablation design** examining 4D supervision, prediction location, and **robot-keypoint versus whole-scene tracking**.
- In the paper's LIBERO-Plus evaluation, ELAN4D improved average success from **53.6% to 67.6%** (π_0) and **73.6% to 78.2%** ($\pi_{0.5}$).

Task-Relevant Representation Learning for VLA Policies

Sept 2025 – Jan 2026

Co-first author · RA, RethinkLab, SJTU · [GuidedVLA](#), [RSS 2026](#) · [Project](#)

Advisors: Prof. Junchi Yan and Prof. Xiaosong Jia.

- Led **codebase implementation and simulation experiments** for GuidedVLA, which explicitly supervises task-relevant factors in a vision–language–action policy.
- Implemented **specialized action-attention heads** for **object grounding**, **temporal skill recognition**, and **depth-based geometric perception** in the π_0 action decoder.
- Integrated a **ControlNet-style residual adapter** for plug-and-play factor supervision while preserving the pretrained policy pathway.
- Evaluated policies on LIBERO-Plus and RoboTwin 2.0; GuidedVLA achieved **75.4%** and **90.6%** average success rates, compared with **68.2%** and **77.4%** for the baseline.
- Optimized the openpi PyTorch training pipeline using **mixed precision** and **torch.compile**.

Selected Projects

RoboMaster University Championship — Team Jiao Loong

Sep 2023–Aug 2025

Vice-Captain & Algorithm Lead (Aug 2024–Aug 2025)

- Directed **schedules and technical strategy** for approximately **15 algorithm members** in SLAM, navigation, and perception, contributing to national titles in 2024 and 2025.
- Developed a **FAST-LIO2-based LiDAR-inertial SLAM** pipeline with **GICP map registration** and **PTP-synchronized LiDAR timing** for robust localization and map alignment during fast robot motion.
- Built a motion planning system combining **kinodynamic A*** and **model predictive control (MPC)** for embedded robot navigation and trajectory tracking under real-world competition constraints.
- Deployed **TensorRT-accelerated YOLO** on **Jetson Orin** and integrated **EKF-based motion prediction** for opponent-state estimation under visual occlusion and high-speed motion in RoboMaster matches.
- Earned the **RM Award (Algorithm)** for distinguished individual contributions to **Sentry SLAM and navigation**, as **one of only three algorithm recipients in 2025**.

Honors & Awards

National Scholarship

2024, 2025

Ministry of Education of China

National Champion, RoboMaster University Championship

2024, 2025

Team Jiao Loong, Shanghai Jiao Tong University

RM Award (Algorithm), RoboMaster University Championship

2025

Individual technical honor; one of three algorithm recipients in 2025

Finalist, Mathematical Contest in Modeling (MCM)

2025

Problem B: environmental dynamic system evolution and multi-objective optimization modeling

Bronze Medal, National Olympiad in Informatics (NOI) Winter Camp

2021

China Computer Federation (CCF)

Platinum Division, USACO

2020

USA Computing Olympiad (USACO)

Technical Skills

Programming: Python, C, C++

Machine Learning & Vision: PyTorch, TensorFlow, OpenCV, TensorRT

Robotics & Development: ROS/ROS 2, Docker, Git, LaTeX

Languages

English; French (DELFB2)

TOEFL iBT: 5.5/6 (ETS comparable total: 110/120), 8 Mar 2026. R 6.0 / L 5.5 / S 6.0 / W 5.0.